

FAZT I4G -16 INTERROGATOR



Description

The FAZT I4-16 optical sensing instrument combines the core technology of the flagship FAZT I4G interrogator together with four 1x4 high-reliability, high-speed, premium performance solid-state optical switches to create a 16 Channels Interrogator. Enclosed in a field deployable stainless-steel casing, the instrument provides peak detection for up to 30 Fiber Bragg Gratings (FBG) based sensors on each of its 16 channels. The instrument can be operated in full spectrum and sensor peak detection modes. The FAZT I4-16 is optimized for both static and dynamic measurements.

Our family of rugged FAZ Technology interrogators is used extensively in marine, railways, roads, energy, civil, geotechnical, industrial, security, medical, and many other commercial applications. It is also the instrument of choice for Research and Development Centers that seek to achieve ultimate precision and repeatability measurements. The industrial grade design scales well in volume production.

Benefits

Ultimate precision, repeatability, and accuracy: The FAZT I4-16 Interrogator returns measurements in 1pm steps across its wavelength range. Using patented FAZ Technology scan-by-scan calibration, the instrument features ultimate 0.1pm absolute precision, superb repeatability of 50fm max, and absolute accuracy of 1pm (Gas Cell and MZI referenced in closed PM circuit). *Ultimate measurements require tight control of all three calibration parameters: precision, accuracy and repeatability.*

Top reliability: The FAZT I4-16 Interrogator is based on a semiconductor tunable laser that has no movable parts, no tunable filters, and no opto-mechanical switches, which delivers top reliability over a broad temperature range to form an integral part of a very rugged and reliable sensing system.

Adaptive to more types of sensors: The integrated electronics and embedded software allow the user to quickly adapt the instrument's performance parameters to fit many different sensor configurations. Designed to monitor with ultimate precision more types of sensors than any other instrument, from narrow bandwidth (40-100pm) FBGs used in high-sensitivity accelerometers, pressure sensors, hydrophones, and microphones, to wide bandwidth (1.5nm) FBGs used in bio-sensing. Hardware implementation of peak tracking algorithms for FBG sensors including programmable gain per sensor, width/height thresholds, and distance to sensor setting for accuracy at long distance. The laser output power (typically ~2 dBm depending on configuration) and high sensitivity receiver yield exceptional optical dynamic range performance over long distances.

Systems and network ready: High speed data acquisition and on-board computer processing make the FAZT I4-16 easy to use and easy to transmit large volumes of data by network connection. Trigger Input support enabling synchronization of multiple optical and electrical sensor systems. The Ethernet port for high speed data transfer is 100Mbit/s, with all interrogator settings programmable over a REST interface. NTP (<10ms timestamps) Time accuracy with optional GPS time (<10µs timestamps).



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Applications include Aerospace, Marine, Railways, Roads, Civil & Geo Engineering, Energy, Industrial, and R&D Labs

Femto Sensing International undertakes a rigorous development process before products release. The company is also firmly committed to continuous improvements after release to ensure performance and reliability to the highest standards, hence, specifications are subject to update without notice.

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PARAMETER	SPECIFICATIONS	NOTES
Wavelengths Range	39.2nm	1529nm to 1568.2nm
Number of Channels	16	
Number of Sensors per Channel (see Note 1)	1 to 30	16ch * 30 = 480 sensors (assuming 1.2nm spacing of FBG sensors)
Wavelength Sample Size / Resolution	1pm	Raw data. Resolution can be further improved with averaging
Wavelength Absolute Accuracy	< 1pm	Gas Cell and MZI referenced in closed loop PM circuit to guarantee long-term performance
Wavelength Precision	< 0.1pm	Defined over an 8 hour test
Wavelength Repeatability (see Note 2)	< 0.05 pm	
Laser Line-Width	20 MHz	Self-heterodyne measured line-width at static wavelength
Laser Output Power per Channel (see Note 3)	+0 to +3dBm	Performance is maintained even with 20dB optical power loss
Scan Frequency / FBG Processing	200Hz	For 16 parallel channels. Also retains 4ch operation at 1/2/4/8 KHz modes of I4G
Scan Frequency (Full Spectrum @ 1pm)	1Hz	1Hz for 16 channels with 1pm resolution
Polarization Switching States	2 States	Switches polarization state every sweep when user enabled
Sensor Range / Distance	0 to 10km	Lead-in cable (0 to 7.5km), FBG section (0 to 3.25km)
Input Voltage and Power Consumption	12V and 25W	Auto-detect 100V to 240V AC with 12V supply block included
Operating Temperature	0 to +55°C	+20°C to +55°C operation range also available
Dimensions (WxDxH) and Weight	324x276x116mm and 4.9kg	Color is RAL-9005 Black
Optical Connection to Sensors	LC/APC	
Certifications and Test Reports	YES	EN55022, EN55024, EN6100, ROHS and REACH Compliant, CE Certified.
Communications Interface	YES	100Mbps Ethernet, REST control interface and multiple data output formats/ports
Supplied Software	YES	FEMTOSENSE configuration tool, LABVIEW interface examples, API Support Document

Note 1: FBGs from 40pm to 1.5nm BW@3dB (FWHM) are supported.

Note 2: Standard deviation measured over a 10 second duration for a stable FBG peak. The FBG FWHM is 100pm and the measured reflected FBG peak power is -15dBm.

Note 3: Dynamic programmable receiver gain per sensor (4 levels covering 12dB of gain) delivering >30dB optical power dynamic range (saturation - minimum detectable power levels) at 200Hz sweep rate.

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